INCH-POUND

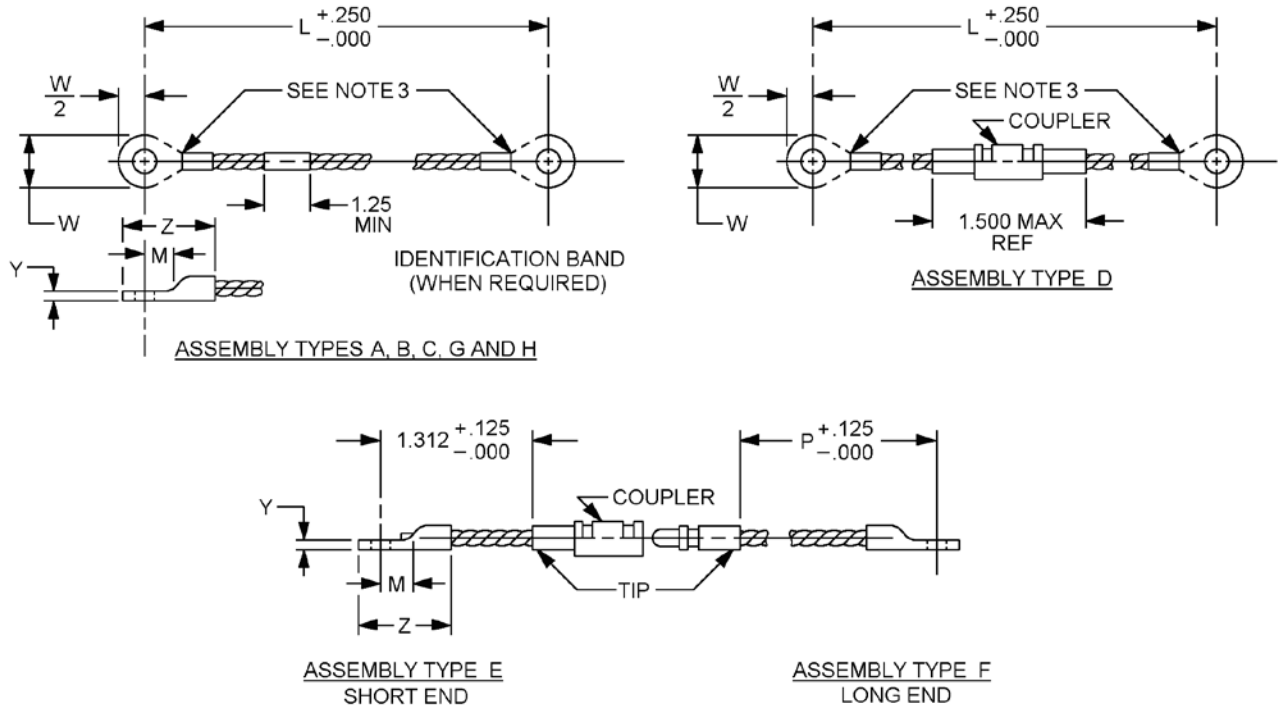
MIL-DTL-83413/8D
w/Amendment 1
30 July 2014
SUPERSEDING
MIL-DTL-83413/8D
17 March 2014

DETAIL SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: TYPE IV JUMPER CABLE ASSEMBLY, LEAD, ELECTRICAL

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.



Inches	mm
.125	3.18
.250	6.35
1.25	31.75
1.312	33.32
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Resistance readings to be taken at the junction of terminal barrels and tongues.

FIGURE 1. Jumper cable assembly.

AMSC N/A FSC 6150

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TABLE I. Jumper assembly types and characteristics.

Jumper assembly type	Construc-tion type	Wire and terminal material	Wire size (see table II)	Available lug sizes (see tables III thru V)	Available cable length, see "L" or "P" dimensions on figure 1	Lightening tested	Fuel Compat-Ible 1/	Method of attaching terminals	Color of ID band	Supersedes
A	Bonding	Copper	12	A thru J	002-999	No	No	Crimp	No band	MS25083-2
B	Current return	Copper	8	K,L,M,N,T,U	002-999	No	No	Crimp	No band	MS25083-4
C 2/	Bonding	Aluminum	10	O,P,Q,R,S (aluminum only)	003-999	No	Yes	Brazed 4/	Clear	MS25083-1 and -7
D 3/	Quick disconnect	Copper	12	A, B, C,D,E, F,G	003-999	No	No	Crimp	No band	MS25083-3
E 3/	Short end of quick disconnect	Copper	12	A, B, C,D,E, F,G	Fixed length	No	No	Crimp	No band	MS25083-3S
F 3/	Long end of quick disconnect	Copper	12	A, B, C,D,E, F,G	002-999	No	No	Crimp	No band	MS25083-3P
G	Bonding	Copper	12	A thru J	003-999	Yes	No	Crimp	Yellow	MS25083-5
H	Current return	Copper	8	K,L,M,N,T,U	003-999	Yes	No	Crimp	Yellow	MS25083-6

1/ Only fuel compatible jumper assemblies may be used within fuel tank and cells.
2/ Do not use outside fuel tanks and cells.
3/ Inactive for Air Force Airborne use after 9 December 1963.
4/ Dip braze terminals in accordance with AWS-C3.7 Specification for Aluminum Brazing. A coating is not required on interior strands of the cable, chemical conversion coating in accordance with MIL-DTL-5541, class 3 shall be applied after brazing. Inert gas welding using an aluminum based solder or other means of termination may be used only if approved by the qualifying activity. Aluminum terminals may be crimped to the cable to facilitate dip brazing. Over crimping shall be avoided.

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TABLE II. Wire characteristics by jumper assembly type.

Jumper assembly type	Material	Stranding no. x AWG	Wire size	Assembly 1/ resistance (max-ohms)	Tensile strength (min-lb)	Copper only	
						Crimping tools 2/	Crimping dies
A, G	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	110	M22520/5-01 3/ or M22520/24-01	M22520/5-100
B, H	Copper	(7 x 95) x 36	8	.0006xL + .00016 Ω	225	SAE-AS5259/1	SAE-AS5259/1 3/
C	Aluminum	37 x 0.0167 inch	10	.00042xL + .00110 Ω	75	Not applicable	Not applicable
D	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	5 to 28 4/	M22520/5-01 M22520/24-01	M22520/5-100
E	Copper	(7 x 37) x 36	12	.00057	110	M22520/5-01 M22520/24-01	M22520/5-100
F	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	110	M22520/5-01 M22520/24-01	M22520/5-100

1/ Where L = length of the cable assembly in inches. Example: For a 24 inch type A assembly, assembly resistance = .00016 ohm/inch x 24 inches + .00034Ω = .00418 ohms (max).
2/ Alternate crimping tools which use the specified dies of this table or qualified class II copper terminal lugs in accordance with SAE-AS7928 (crimped with the tools and dies used to obtain the class II terminal lug approval), may be used to facilitate high volume production of copper jumpers. However, either of these options must be pre-approved by the qualifying activity before qualification tests are initiated on samples produced using these termination options.
3/ Application: M22520/5-01, /5-100 for use on insulated terminals. M22520/24-01 and SAE-AS5259/1 for use on uninsulated terminals.
4/ Tensile strength for D jumper assemblies is 5 to 28 lbs, however for the separate assembly ends: E = 110 lbs and F=110 lbs.

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TABLE III. Available lug sizes.
A,D,E,F & G type jumper assembly. 1/ 2/ 3/ 4/ 8/

Code letter lug size designation	For stud size	12 AWG Copper terminals (All types except type C)	
		SAE-AS25036 Insulated	SAE-AS20659 Uninsulated
A	.138 (3.51)	-111	…
F	.138 (3.51)	…	-165
I	.164 (4.17)	-156	…
B	.190 (4.83)	-112	…
G	.190 (4.83)	…	-105
C	.250 (6.35)	-157	…
D	.312 (7.94)	-113	…
H 6/	.312 (7.94)	…	-106
E	.375 (9.53)	-114 5/	…
J 7/	.375 (9.53)	…	-128

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ±.005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.
4/ Lightning current tests applicable to type G and H assemblies only.
5/ Size #10 & #12 AWG wire only.
6/ Lug size designators D & H correspond to lug size D in Rev B, but were broken out separately for clarification.
7/ Lug size designators E & J correspond to lug size E in Rev B, but were broken out separately for clarification.
8/ For supersession information see table VI.

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TABLE IV. Available lug sizes.
B & H type jumper assembly. 1/ 2/ 3/ 4/ 8/

Code letter lug size designation	For stud size	8 AWG Copper terminals (All types except type C)	
		SAE-AS25036 Insulated	SAE-AS20659 Uninsulated
T	.250 (6.35)	-116	…
U	.250 (6.35)	…	-141
K 6/	.312 (7.94)	-117 5/	…
L 6/	.312 (7.94)	…	-108 5/
M 7/	.375 (9.53)	-118	…
N 7/	.375 (9.53)	…	-129 5/
W	.164 (4.17)	…	-140
V	.190 (4.83)	…	-107
Z	.190 (4.83)	-115	…

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ±.005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.
4/ Lightning current tests applicable to type G and H assemblies only.
5/ Size #8 AWG wire only.
6/ Lug size designators K, & L, correspond to lug size D in Rev B, but were broken out separately for clarification.
7/ Lug size designators M, & N, correspond to lug size E in Rev B, but were broken out separately for clarification.
8/ For supersession information see table VII.

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TABLE V. Available lug sizes.
C type jumper assembly. 1/ 2/ 3/ 6/

Code letter lug size designation	For stud size	Aluminum terminal				
		Stud Hole ID	M min	W	Y	Z max
O	.138 (3.51)	.152 (3.86)	.218 (5.54)	.406 (10.31)	.083 (2.11)	1.172 (29.77)
		.142 (3.61)		.313 (7.95)	.037 (.939)	
P	.190 (4.83)	.203 (5.16)	.250 (6.35)			
		.193 (4.90)				
Q	.25 (6.35)	.285 (7.24)	.281 (7.14)	.540 (13.72)		1.300 (33.02)
		.260 (6.60)		.450 (11.43)		
R 4/	.312 (7.94)	.343 (8.71)	.329 (8.36)	.741 (18.82)	.090 (2.29)	1.390 (35.31)
		.320 (8.13)		.531 (13.49)	.038 (.965)	
S 5/	.375 (9.53)	.385 (9.78)	.343 (8.71)			1.626 (41.30)
		.410 (10.41)				

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ±.005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.
4/ Lug size designators R, correspond to lug size D in Rev B, but were broken out separately for clarification.
5/ Lug size designators S, correspond to lug size E in Rev B, but were broken out separately for clarification.
6/ For supersession information see table VIII.

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REQUIREMENTS:

Material:

For jumper types A,D,E,F, and G: Soft or drawn and annealed tin coated copper wire per ASTM-B172. Rope-lay-bunched member stranding consisting of 7 bunches of 37 individual strands of 36 AWG equivalent solid copper wire (259 individual wire strands with total approximate circular mill area 6475 with 12 AWG total equivalent).

Aluminum wire (type C assembly): Size 10, 37 strands of 0.0167 inch, electrical grade, aluminum alloy, 1350 strands in accordance with ASTM-B230/B230M, concentric stranded, length of lay (pitch) .75 to 1.25 inch.

Jumper types B and H: Soft or drawn and annealed. Tin coated copper wire per ASTM-B172. Rope-lay-bunched member stranding consisting of 7 bunches of 95 individual strands of 36 AWG equivalent solid copper wire (665 individual wire strands with total approximate circular mill area 16625 with 8 AWG total equivalent).

Coupler (assembly types “D”, “E” and “F” only): In accordance with MIL-DTL-6852.

Identification band (types C, G and H only): Polyolefin, heat shrinkable in accordance with SAE-AMS-DTL-23053/5. Color specified in table I.

Assembly resistance: See table II.

Tensile strength: See table II. Perform after the flexibility test in accordance with MIL-DTL-83413, group II qualification sequence. Omit assembly resistance as a tensile strength post test.

Crimping tools and crimping dies: See table II.

Marking (types C, G and H only): Part or Identifying Number (PIN) shall be marked on the identification band. If the identification band is too short to accommodate the entire PIN on a single line, the PIN may be put on two lines or both sides of the identification band. Assembly type “C” parts shall be marked with an ink or fluid which is compatible with fuel.

Salt spray: 48 hours in accordance with test procedure EIA/ECA-364-26.

Altitude-low temperature: Not applicable.

Dust: Not applicable.

Ozone exposure: Not applicable.

Lightning current: Not applicable to types A, B, C, D, E and F.

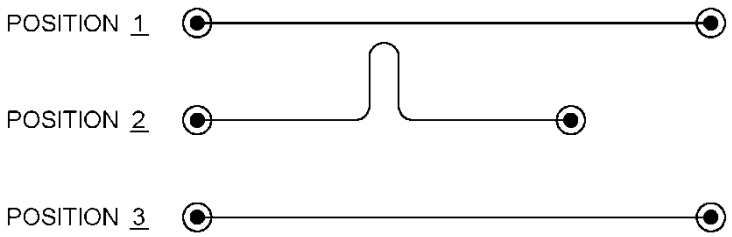
Dimensions and configuration: See figure 2.

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Flexibility: For type “C” aluminum assemblies, change the MIL-DTL-83413 test to the following: Twist and flex: prior to the tensile strength test of MIL-DTL-83413.

Test No. 1 – Twist test 2 half twists (360º) in a direction to tighten the wire weave. This is for clarification and to accommodate the short 3 inch C type jumper.

Test No. 2 – Flex 20 times.



NOTE: For copper assemblies the flexibility test in accordance with MIL-DTL-83413 shall be performed at a rate of 30-65 cycles/minute.

FIGURE 2. Twist and flex test.

Qualification required: The activity responsible for the qualified products list for this specification sheet is the DLA Land and Maritime, Columbus, DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

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TABLE VI. Supersession table.
A,D,E,F & G type jumper assemblies.

Superseded LUG Code (Rev B)	Preferred LUG Code
A	A
F	F
I	I
B	B
G	G
C	C
D	D
D	H
E	E
E	J

TABLE VII. Supersession table.
B & H type jumper assembly.

Superseded LUG Code (Rev B)	Preferred LUG Code
D	K
D	L
E	M
E	N
----	T (NEW)
----	U (NEW)

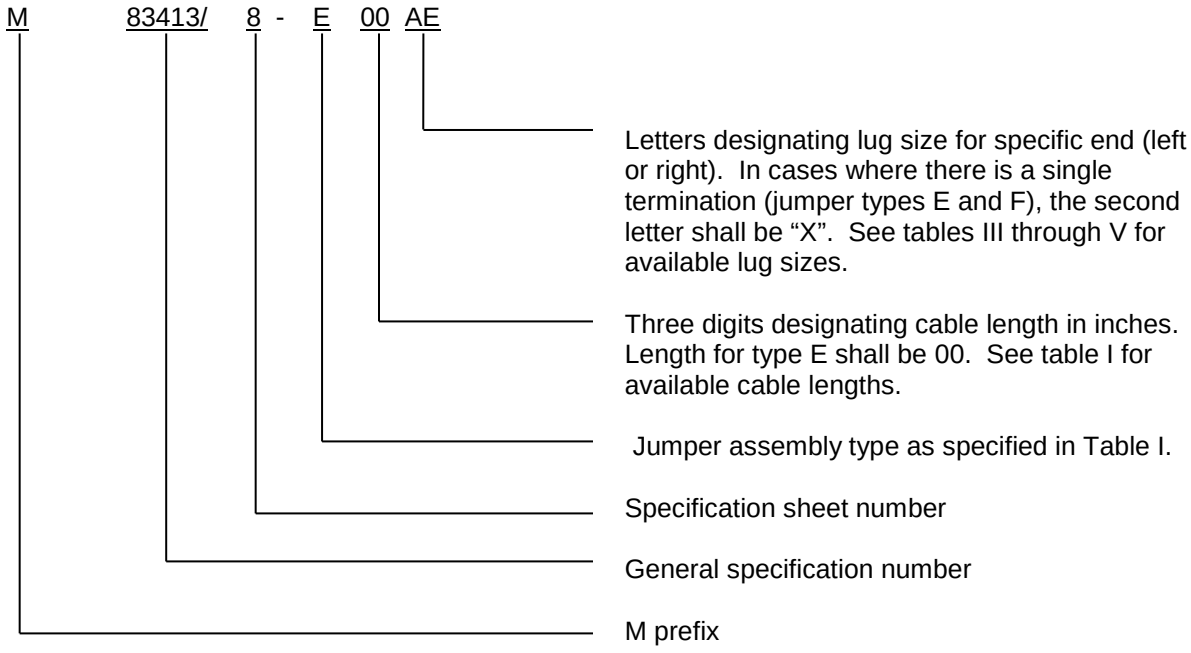
TABLE VIII. Supersession table.
C type jumper assembly.

Superseded LUG Code (Rev B)	Preferred LUG Code
A	O
B	P
C	Q
D	R
E	S
F	No replacement 1/
G	No replacement 1/

1/ AS20659 is Copper, Uninsulated. This jumper is Aluminum.

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Example of PIN: M83413/8-E00AE



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Referenced documents. In addition to MIL-DTL-83413, this document references the following:

MIL-DTL-5541	SAE-AS25036
SAE-AS5259/1	SAE-AS20659
ASTM-B230/B230M	SAE-AMS-DTL-23053/5
EIA/ECA-364-26	AWS-C3.7
SAE-AS7928	ASTM-B172
MIL-DTL-6852	SAE-AS5259/1

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6150-2014-001)

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force - 71, 99
DLA -GS

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